Predicting Port Congestion & Shipment Delays Using AI

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Assignment: Aiqon Take-Home Challenge (Round 2)

Target Model: Random Forest Classifier

1. Executive Summary & Problem Framing

Port congestion is a primary driver of supply chain instability, triggering compounding delays, escalating demurrage costs, and operational friction across ocean freight networks. This technical assignment delivers an end-to-end machine learning system capable of anticipating port bottlenecks before vessel arrival.

In this implementation, **port congestion** is framed as a binary risk condition where expected total vessel delays exceed a 4.0-hour turnaround threshold relative to standard operational baselines. Rather than operating purely as a black-box predictor, the primary focus of this design is double-fold: generating accurate congestion risk probabilities (confidence scores) and providing granular, human-readable explanations of key operational drivers.

2. Data Engineering & Synthetic Simulation Architecture

To capture real-world operational dynamics without raw AIS pipeline noise, a 300-record synthetic dataset was structured incorporating key marine terminal variables:

- **Vessel Arrival Density:** Daily inbound traffic rate (range: 5 to 35 vessels/day).
- **Port Handling Capacity:** Discrete terminal processing throughput limits (15, 20, 25, 30 ships/day).
- **Queue Length:** Active vessels waiting at anchor prior to berth assignment (0 to 15 vessels).
- **Weather Condition:** Categorical environmental disruption risk (Clear, Moderate, Severe).
- **Historical Delay Average:** Rolling 30-day mean turnaround deviation (0.5 to 6.0 hours).

Ground truth congestion labels (`is_congested`) were modeled by evaluating terminal saturation metrics: combining vessel arrival deficits, anchor queue lengths, and non-linear weather risk multipliers.

> **Ground Truth Logic Formula:**
> $\text{Congestion Score} = 2 \cdot I(\text{Queue} > 8) + 3 \cdot I(\text{Arrival} > \text{Capacity}) + \text{Weather Risk} + 1.5 \cdot I(\text{Hist Delay} > 3.0) + \varepsilon$
> Shipments with scores exceeding 4.5 are flagged as high risk (1).

3. Machine Learning Architecture & Model Training

The prediction engine utilizes a `RandomForestClassifier` composed of 100 decision trees trained on an 80/20 train-test data split. Categorical variables were processed using one-hot encoding.

Model Evaluation Metrics

The ensemble classifier demonstrated strong performance across standard evaluation metrics, proving effective at distinguishing normal terminal throughput from severe bottleneck scenarios.

Evaluation Metric	Score / Value	Operational Interpretation
Classification Accuracy	90.00%	High overall accuracy across test samples.
ROC-AUC Score	0.9506	Excellent separation between congested vs clear states.
Precision (Class 1)	0.8600	86% of flagged congestion alerts represent genuine delays.
Recall (Class 1)	0.9300	93% of actual congestion incidents successfully captured.

4. Model Interpretability & Feature Importances

To ensure predictions are actionable for terminal operators and logistics managers, feature importances were extracted directly from tree node impurities:

Feature Name	Gini Importance Score	Relative Contribution
`vessel_arrival_density`	0.3301	Primary bottleneck driver (33.0%)
`queue_length`	0.2141	Secondary waiting risk (21.4%)
`historical_delay_avg`	0.1717	Baseline delay trend (17.2%)
`port_capacity`	0.1228	Terminal handling limit (12.3%)
`weather_condition_Severe`	0.1110	Environmental hazard impact (11.1%)
`weather_condition_Moderate`	0.0503	Minor environmental impact (5.0%)

5. Live Inference & Operational Decision-Making

The inference module (`src/predict.py`) maps prediction probabilities directly into confidence percentages and attaches explicit key drivers to inform supply chain planning.

Sample System Output Test

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Input Parameters: - Vessel Arrival Density: 28 vessels/day - Port Capacity: 20 vessels/day - Anchor Queue Length: 10 waiting ships - Weather Condition: Severe - Historical Delay Avg: 4.5 hours System Output: Congestion Flag: True Congestion Probability: 0.9800 Confidence Score: 98.0% Key Contributing Factors: - High port queue length (10 ships waiting) - Arrival density (28) exceeds port capacity (20) - Adverse weather condition (Severe) - High historical port delay average (4.5 hrs)
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6. Real-World Limitations & Scalability Considerations

While this solution provides a clear proof-of-concept, real-world deployment across global ocean logistics requires addressing specific operational challenges:

- **Dynamic AIS Data Stream Integration:** Real-world implementations require real-time AIS vessel speed, draft, and terrestrial positioning telemetry to estimate berth arrival times dynamically.
- **Seasonal Traffic Volatility:** Port demand spikes dramatically during peak quarter shipping seasons (e.g., Q3/Q4 holiday rushes), requiring time-series LSTM or Transformer architectures.
- **Multi-Port Cascading Delays:** A bottleneck at a transshipment hub like Singapore or Rotterdam propagates downstream delays; future iterations should incorporate graph network models.

7. Conclusion

The implemented Random Forest framework achieves a 90% accuracy and 0.9506 ROC-AUC score while maintaining interpretable outputs. By combining confidence scores with specific operational drivers, logistics teams can make informed rerouting and scheduling decisions.